
The Coordinated Attack on a Warehouse Floor

Common Knowledge as an Action Precondition,
a Lossy Channel that Cannot Supply It, and a Signal that Can

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Two robots must lift one load together, one under each end. A work order names which of two stands, and only one robot can read it. Since an end raised alone tips the load, each robot raises its end only if it knows the other raises the other, and the weakest condition satisfying both is common knowledge of the stand. The report states that condition as the precondition of an ontic action in a dynamic epistemic planning domain, and asks which channel can establish it.

A message over a channel that can lose it is given as an event model with three events. We show that the product update with such a message preserves S5, and that it decreases the distance from the designated worlds to a counterexample by at most one, so that each message adds at most one level of mutual knowledge and no finite sequence of them yields common knowledge. On the floor without a beacon this proves that no policy exists, for messages of any number; the planner, on a bounded protocol, exhausts its space at depth five. A stack light visible only from two viewpoints yields common knowledge in one update, and only when both robots stand at their viewpoints. When a move is witnessed only by the mover, the same signal yields one level of mutual knowledge, as a message does, and the planner has the robots sight each other first. All four floors were executed in Gazebo through ePlanSys. Over the radio, four delivered messages reached E^4 and the executor refused the lift; with the beacon, the policy lifted the load from a joint start, with arrivals 0.40s apart. With positions unannounced, the robots sighted each other with their own lasers and lifted; with crates across the sight line, north saw the signal, south could not know that it had, and the executor refused the lift at E^1 .

Contents

1	Scope	2
2	Preliminaries	2
3	The domain	3
3.1	The lossy message	4
3.2	The conditional announcement	5
4	The radio cannot supply the precondition	5
4.1	The ladder	6
5	The beacon supplies it in one update	7
6	Positions that are not announced	8
7	Planning results	9
8	Grounding on the floor	9
9	Runs	11
9.1	The radio floor	11
9.2	The beacon floor	11
9.3	The positions floors	12
10	Defects	14
11	What the runs establish	15
12	Directions for the next iteration	15

1 Scope

The companion reports execute epistemic policies whose preconditions are individual knowledge: a carrier crosses a bay it knows to be open, a robot reports a site whose state it knows. This report concerns a precondition of a different kind. The action is joint, and its precondition is common knowledge of a group.

The problem is the coordinated attack of Gray [5] and of Akkoyunlu, Ekanadham and Huber [1], in the form Halpern and Moses gave it [6]: two parties must act simultaneously or not at all, communicate over a channel that may lose messages, and cannot coordinate. It is restated here as a warehouse task on the floor of the pass-through study, and the result is obtained in three ways: by proof over the event models of the domain, by exhaustive search of the planner, and by execution on simulated robots.

Four claims are advanced.

1. A message over a lossy channel, as an event model, preserves S5 and adds at most one level of mutual knowledge per message, whether or not it is delivered (Propositions 1 and 2). On the floor without a beacon no sequence of actions of any length makes the lift applicable (Corollary 1).
2. A signal observed from viewpoints that are common knowledge yields common knowledge in one update, and the same signal observed by one robot yields no mutual knowledge at all (Propositions 3 and 4).
3. The planner agrees with both results: it exhausts the radio floor and returns a policy of depth five on the beacon floor that does not use the radio.
4. Both floors were executed in simulation with the precondition checked by the executor against the model, and the depth of mutual knowledge computed from the live model after every update agreed with the offline trace at every step.

Four bounds are stated at the outset. Every message in the runs is delivered: only delivery is designated, and the argument is that delivery does not suffice. The lift is drawn by moving the load in the simulator once both robots are under it. The positions of the robots are common knowledge by the argument of Section 3, and are not learned. And the planner's search on the radio floor is bounded to one message per level of nesting; the general statement is Corollary 1.

2 Preliminaries

Definition 1 (Epistemic model). Let $\mathcal{A}$ be a finite set of agents and P a finite set of atoms. An *epistemic model* is $\mathcal{M} = (W, R, V, W_d)$ with W a finite set of worlds, $R : \mathcal{A} \rightarrow 2^{W \times W}$, $V : W \rightarrow 2^P$, and $\emptyset \neq W_d \subseteq W$ the designated worlds. $\mathcal{M}$ is S5 when every R_a is an equivalence. A formula holds at $\mathcal{M}$ when it holds at every designated world.

Definition 2 (Group modalities). For $G \subseteq \mathcal{A}$ let $R_G = \bigcup_{a \in G} R_a$ and R_G^+ its transitive closure. Then

$$\mathcal{M}, w \models E_G \varphi \iff \forall v (w R_G v \Rightarrow \mathcal{M}, v \models \varphi), \quad E_G^{k+1} \varphi = E_G E_G^k \varphi, \quad E_G^0 \varphi = \varphi,$$

$$\mathcal{M}, w \models C_G \varphi \iff \forall v (w R_G^+ v \Rightarrow \mathcal{M}, v \models \varphi),$$

so that $C_G \varphi$ holds exactly when $E_G^k \varphi$ holds for every $k \geq 1$.

Definition 3 (Distance and depth). For a model $\mathcal{M}$ with $\mathcal{M} \models \varphi$, let $\text{dist}_{\mathcal{M}}(\varphi)$ be the least m such that some designated world reaches, by an R_G -path of length m , a world where φ fails; ∞ if there is none. The *depth* of φ is $\text{depth}_{\mathcal{M}}(\varphi) = \text{dist}_{\mathcal{M}}(\varphi) - 1$.

Lemma 1. If $\mathcal{M}$ is S5 and $\mathcal{M} \models \varphi$, then for $k \geq 1$, $\mathcal{M} \models E_G^k \varphi$ iff $k \leq \text{depth}_{\mathcal{M}}(\varphi)$, and $\mathcal{M} \models C_G \varphi$ iff $\text{dist}_{\mathcal{M}}(\varphi) = \infty$.

Proof. $\mathcal{M}, w \models E_G^k \varphi$ iff φ holds at every world reached from w by an R_G -path of length exactly k . Each R_a is reflexive, so a path of length $m \leq k$ extends to one of length k ending at the same world, and the worlds reached in exactly k steps are those reached in at most k . Hence $E_G^k \varphi$ fails at some designated world iff a counterexample lies within distance k of it, that is iff $\text{dist}_{\mathcal{M}}(\varphi) \leq k$. The case of C_G is the union over k . $\square$

Lemma 1 is what makes the depth computable by a breadth-first walk from the designated worlds, which is how the trace tool and the knowledge view of the runs compute it, and why the walk also yields a witness: the chain of agents along the shortest path to a counterexample.

Definition 4 (Event model with observability types). An *event model* is $\mathcal{E} = (E, T, Q, \text{pre}, \text{post}, E_d)$ with E a finite set of events, T a finite set of observability types, $Q : T \rightarrow 2^{E \times E}$, $\text{pre} : E \rightarrow \mathcal{L}$, $\text{post} : E \rightarrow (P \rightarrow \mathcal{L})$ and $E_d \subseteq E$ the designated events. An *observability assignment* gives each agent a a type $\tau_a \in T$ for the whole model: the type whose observability condition holds at every designated world. This is how plank's product update evaluates the conditions, and Aletheia's after it.

Remark 1. The textbook reading of a conditional observability gives each agent a type $\tau_a(w)$ at each world, by the condition at w . The two readings agree whenever the conditions are common knowledge, which is the case in every model of Sections 4 and 5: the lossy message has fixed types, and the signal's condition is a position, which the public move to a viewpoint makes common knowledge. They disagree when it is not, and under the reading the tools implement, whether an agent observed an action is common knowledge by construction. Section 6 needs it not to be, and says so in the event model.

Definition 5 (Product update). $\mathcal{M} \otimes \mathcal{E} = (W', R', V', W'_d)$ with $W' = \{(w, e) : \mathcal{M}, w \models \text{pre}(e)\}$,

$$(w, e) R'_a (v, f) \iff w R_a v \text{ and } (e, f) \in Q(\tau_a),$$

$V'(w, e) = \{p : \mathcal{M}, w \models \text{post}(e)(p)\}$, with $\text{post}(e)(p) = p$ where unspecified, and $W'_d = \{(w, e) \in W' : w \in W_d, e \in E_d\}$. $\mathcal{E}$ is *applicable* in $\mathcal{M}$ when every designated world has some designated event whose precondition holds there.

Fact 1 (Halpern and Moses [6]; see also [4], Ch. 6 and 11). In a system where an action must be performed simultaneously by every agent of a group G or by none, the action is performed only at points where it is common knowledge in G that it is being performed. If communication is not guaranteed, common knowledge of any fact not common knowledge initially is never attained.

The domain takes the first half of Fact 1 as its modelling assumption and states it as a precondition. The second half is not assumed: Section 4 derives it for the event model by which the domain represents a message.

3 The domain

Agents $\mathcal{A} = G = \{\text{south}, \text{north}\}$, stands $S = \{s_1, s_2\}$. The initial model has two worlds, both designated, with $\text{job}(s_k)$ true in w_k only, and both relations total:

$$s \models C_G(\text{job}(s_1) \oplus \text{job}(s_2)), \quad s \models \neg K w_a \text{job}(s_k) \quad \text{for every } a \in G, k \in \{1, 2\}.$$

The facts $\text{reads-order}(\text{south})$ and $\neg \text{reads-order}(\text{north})$ are common knowledge, and so is the floor: whether it has a beacon, beacon , is the one line in which the two instances differ. The goal is *lifted*, and its epistemic content is the precondition of *lift*.

Table 1: The actions of the domain. ψ_k is the content of the k -th message: $\psi_1 = \text{job}(s)$, $\psi_{k+1} = K_i\psi_k$ with the sender and receiver alternating.

action	type	events	observers
$\text{read-order}(i, s)$	semi-private sensing	$e_h: \text{job}(s); e_e: \neg \text{job}(s)$; both with $\neg \text{at-view}(i) \wedge \neg K_w i \text{job}(s)$	i fully, the other partially
$\text{tell}, \text{ack}, \text{ack2}, \text{ack3}(i, j, s)$	lossy message	$d, l: K_i\psi_k \wedge \neg p_k$, post p_k ; $n: \top$	i sender, j receiver
$\text{go-view}(i)$	public ontic	$\text{beacon} \wedge \neg \text{at-view}(i)$, post $\text{at-view}(i)$	everyone
$\text{signal}(i, s)$	conditional announcement	$\sigma: \text{at-view}(i) \wedge K_i \text{job}(s)$; $n: \top$	at a viewpoint fully, elsewhere obviously
$\text{lift}(s)$	public ontic	$\text{job}(s) \wedge C_G \text{job}(s)$, post lifted	everyone

Why positions are common knowledge. go-view is public. Both robots execute the same policy, and before the branch on the work order every leaf prescribes the same actions, so where each robot is follows from the policy and the public history and needs no observation. The radio carries the work order; it does not carry positions.

Why the order is read before the viewpoint. The terminal is not a viewpoint, so a robot at its viewpoint must leave it to read. Sensing events carry no postconditions in the action-type library, so the domain cannot state that reading takes the reader away; it requires $\neg \text{at-view}(i)$ instead. Section 10 records the run that motivated it.

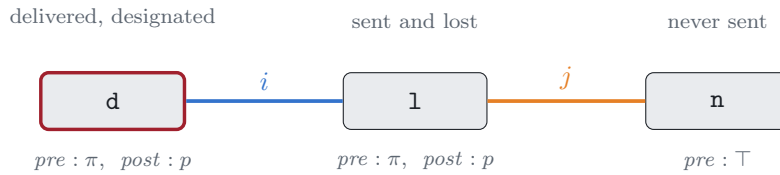
3.1 The lossy message

Definition 6 (Lossy message). For agents $i \neq j$, a formula π and an atom p , the event model $L(i, j, \pi, p)$ has events d (delivered), l (lost) and n (never sent), $\text{pre}(d) = \text{pre}(l) = \pi$, $\text{pre}(n) = \top$, $\text{post}(d) = \text{post}(l) = \{p \mapsto \top\}$, $\text{post}(n)$ trivial, $E_d = \{d\}$, and types

$$Q(\text{Sender}) = \{d, l\}^2 \cup \{(n, n)\}, \quad Q(\text{Receiver}) = \{(d, d)\} \cup \{l, n\}^2,$$

with $\tau_i \equiv \text{Sender}$ and $\tau_j \equiv \text{Receiver}$.

The sender knows it sent and cannot tell whether the message arrived; the receiver knows whether something arrived and, when nothing did, cannot tell a loss from silence. The intermediate action-type library has no such type, and it is given in a library of its own (Listing 1).

**Figure 1:** The lossy message $L(i, j, \pi, p)$. Reflexive edges are omitted. The sender i relates d and l ; the receiver j relates l and n . Both relations are equivalences.

```
(:action-type lossy-message
:events (?del ?lost ?nil)
:observability-types (Sender Receiver Oblivious)
:relations (Sender (:and (?del ?del) (?del ?lost) (?lost ?del) (?lost ?lost) (?nil ?nil))
Receiver (:and (?del ?del) (?lost ?lost) (?lost ?nil) (?nil ?lost) (?nil ?nil))
Oblivious (:forall (?e - event) (?e ?nil)) )
:designated (?del)
:conditions (?nil (:trivial-event)))
```

Listing 1: The action type, in `epddl/lossy.epddl`.

3.2 The conditional announcement

Definition 7 (Signal). For an agent i and a formula χ , $S(i, \chi)$ has events σ and n , $pre(\sigma) = at-view(i) \wedge \chi$, $pre(n) = \top$, both postconditions trivial, $E_d = \{\sigma\}$, and types $Q(\text{Fully}) = \{(\sigma, \sigma), (n, n)\}$ and $Q(\text{Oblivious}) = \{(\sigma, n), (n, n)\}$, with $\tau_a = \text{Fully}$ if $at-view(a)$ holds at the designated worlds and Oblivious otherwise.

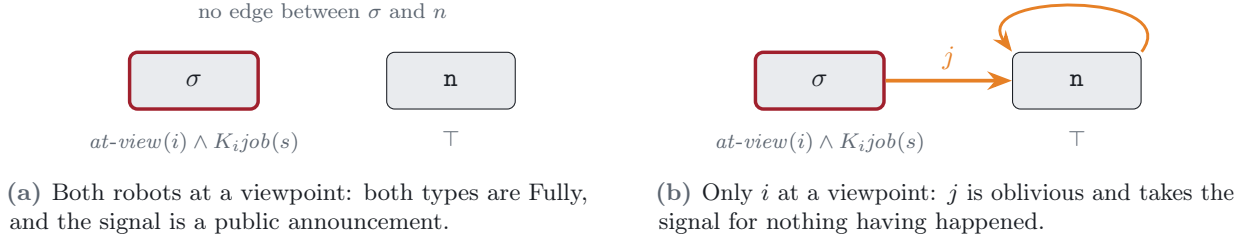


Figure 2: The signal $S(i, K_ijob(s))$ under the two observability assignments that arise.

4 The radio cannot supply the precondition

Proposition 1 (S5 is preserved). If $\mathcal{M}$ is S5, so is $\mathcal{M} \otimes L(i, j, \pi, p)$.

Proof. R'_a is the restriction to W' of the relation $\{((w, e), (v, f)) : wR_av, (e, f) \in Q(\tau_a)\}$ on $W \times E$, which is an equivalence as a product of the equivalences R_a and $Q(\tau_a)$; here τ_a does not depend on the world. The restriction of an equivalence to a subset is an equivalence. $\square$

The third event is not needed for Proposition 1: a message with events d and n alone, the sender relating them and the receiver not, also preserves S5. It is needed for the sender to know that it sent. In the two-event model the sender considers possible a world in which it did not send, and K_ip fails at the designated world; with l the sender's class at (w, d) is $\{(w, d), (w, l)\}$, at both of which p holds.

Proposition 2 (At most one level per message). Let $G = \{i, j\}$, let $\mathcal{M}$ be S5 with $\mathcal{M} \models \varphi$, let φ not mention p , and let π be such that $\mathcal{M}, w \models \pi$ and wR_iv imply $\mathcal{M}, v \models \pi$. If $L = L(i, j, \pi, p)$ is applicable in $\mathcal{M}$, then

$$\text{dist}_{\mathcal{M} \otimes L}(\varphi) \leq \text{dist}_{\mathcal{M}}(\varphi) + 1.$$

Proof. If $\text{dist}_{\mathcal{M}}(\varphi) = \infty$ there is nothing to show. Otherwise let $w \in W_d$ and $w = v_0 \xrightarrow{a_1} v_1 \xrightarrow{a_2} \dots \xrightarrow{a_m} v_m$ be an R_G -path with $m = \text{dist}_{\mathcal{M}}(\varphi) \geq 1$ and $\mathcal{M}, v_m \not\models \varphi$. Since L is applicable, $(w, d) \in W'_d$. Since $pre(n) = \top$, $(v, n) \in W'$ for every $v \in W$, and since $(n, n) \in Q(\tau)$ for both types, vR_au implies $(v, n)R'_a(u, n)$: the n -copies form a copy of $\mathcal{M}$. As $post(n)$ is trivial and φ does not mention p , (v_m, n) falsifies φ . Two cases.

$a_1 = j$. $(v_0, l) \in W'$ since $pre(l) = pre(d)$. Then $(v_0, d) R'_i (v_0, l)$ by reflexivity and $(d, l) \in Q(\text{Sender})$, and $(v_0, l) R'_j (v_1, n)$ by $v_0 R_j v_1$ and $(l, n) \in Q(\text{Receiver})$.

$a_1 = i$. $\mathcal{M}, v_1 \models \pi$ by the hypothesis on π , so $(v_1, l) \in W'$. Then $(v_0, d) R'_i (v_1, l)$ by $v_0 R_i v_1$ and $(d, l) \in Q(\text{Sender})$, and $(v_1, l) R'_j (v_1, n)$ by reflexivity and $(l, n) \in Q(\text{Receiver})$.

In both cases two steps reach (v_1, n) , and $m - 1$ further steps along the n -copy of the path reach (v_m, n) : a path of length $m + 1$ from a designated world to a counterexample. $\square$

The hypothesis on π holds for the four messages of the domain: $\pi = K_i\psi_k \wedge \neg p_k$, where $K_i\psi_k$ is preserved along R_i by positive introspection and p_k is false in every world until the k -th message. The proof does not use that d is the event that occurred: the witness path runs through l and n , so the bound is the same whether the message is delivered or lost, and it is attained in the runs although every message was delivered (Table 2).

Corollary 1 (No finite exchange yields common knowledge). Under the hypotheses of Proposition 2, if $\mathcal{M} \not\models C_G\varphi$ then $\mathcal{M} \otimes L_1 \otimes \dots \otimes L_k \not\models C_G\varphi$ for every finite sequence of lossy messages, and $\text{depth}(\varphi) \leq \text{depth}_{\mathcal{M}}(\varphi) + k$. On the radio floor no sequence of applicable actions makes $\text{lift}(s)$ applicable for either s , and no policy for lifted exists, of any depth.

Proof. The first part is Proposition 2 and Lemma 1, by induction on k ; Proposition 1 keeps each intermediate model S5. On the radio floor *beacon* is false, so *go-view* and *signal* are never applicable. A message requires $K_{\text{south}}\text{job}(s)$ or a statement nested over it, which fails until the order is read, and *read-order* is applicable at most once, since afterwards $Kw_{\text{south}}\text{job}(s)$ holds for both s . Hence every execution is *read-order* followed by lossy messages. Before the order is read, $\text{job}(s)$ fails at one of the two designated worlds, so $\text{lift}(s)$ is not applicable. After it, on the branch where the order names s , the designated world is (w_s, e) for the observed event e , at which $\text{job}(s)$ holds, and north, which observes the sensing partially, relates it to the copy of the other world: $\text{dist}(\text{job}(s)) = 1$. By the first part the distance stays finite after every message, so by Lemma 1 $C_G\text{job}(s)$ never holds, and the precondition of $\text{lift}(s)$ never does. $\square$

Corollary 1 is the second half of Fact 1, obtained for this event model by a direct argument over the product update, and it is stronger than what the search establishes. The planner cannot exhaust a space in which messages repeat: each repetition yields a model no earlier one is bisimilar to. The domain therefore lets each message go out once, and the planner proves the bounded statement only (Section 7).

4.1 The ladder

The bound of Proposition 2 is attained. Table 2 gives the models after each action of the protocol on the branch where the order names s_1 , computed by the product update of the trace tool from the task plank grounds; the world counts are those the epistemic state logged during the recorded run.

Table 2: The protocol on the radio floor. Worlds are those reachable from the designated world; the chain is the sequence of agents along the shortest path to a world where $\text{job}(s_1)$ fails.

action	$ W $	depth of $\text{job}(s_1)$	chain
$\text{read-order}(\text{south}, s_1) \rightarrow e_h$	2	E^0	north
$\text{tell}(\text{south}, \text{north}, s_1)$	4	E^1	south, north
$\text{ack}(\text{north}, \text{south}, s_1)$	6	E^2	north, south, north
$\text{ack2}(\text{south}, \text{north}, s_1)$	8	E^3	south, north, south, north
$\text{ack3}(\text{north}, \text{south}, s_1)$	10	E^4	north, south, north, south, north
$\text{lift}(s_1)$	not applicable: $C_G\text{job}(s_1)$ fails		

Figure 3 draws the model after the fourth message. A world is labelled by what happened to each message along its history: delivered (D), lost (L) or never sent ($\cdot$). Each step of the chain from the actual world undoes one message: north, the sender of ack3 , cannot exclude that it was lost; south, in that world, cannot exclude that ack2 was lost; and so on, until a world in which nothing was sent and north, which never heard, cannot exclude s_2 .

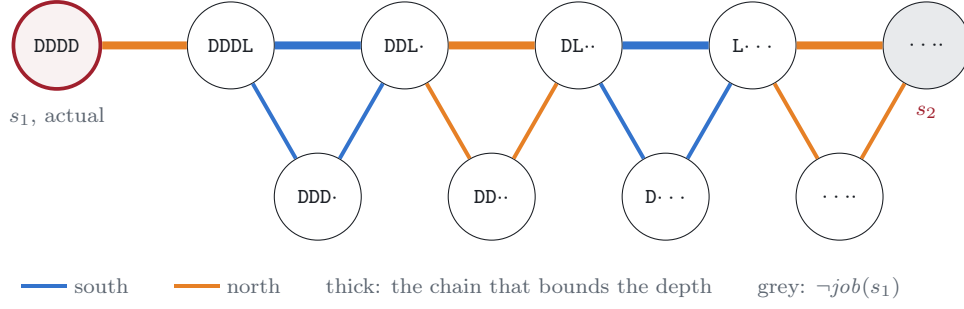


Figure 3: The model after *tell*, *ack*, *ack2* and *ack3*, all delivered: ten worlds. The chain from the actual world to $\neg\text{job}(s_1)$ has length five, so $E_G^4\text{job}(s_1)$ holds and $E_G^5\text{job}(s_1)$ does not (Lemma 1). Reflexive edges and the transitive closure of each class are omitted.

5 The beacon supplies it in one update

Proposition 3 (A signal seen from common viewpoints). Let $\mathcal{M}$ be S5 with *at-view*(*a*) true at every world of $\mathcal{M}$ for every $a \in G$. If $S = S(i, K_i\chi)$ is applicable in $\mathcal{M}$, then $\mathcal{M} \otimes S \models C_G\chi$.

Proof. Every type is Fully, so $(w, e)R'_a(v, f)$ implies $e = f$. Every world R_G^+ -reachable from a designated (w, σ) is therefore of the form (v, σ) , and $(v, \sigma) \in W'$ requires $\mathcal{M}, v \models K_i\chi$, which by reflexivity of R_i gives $\mathcal{M}, v \models \chi$. As *post*(σ) is trivial, χ holds at every reachable world, which is $C_G\chi$. $\square$

Proposition 4 (A signal seen by one robot). Let *at-view*(*j*) be false at every world of $\mathcal{M}$. Then for every formula θ built from atoms, $\mathcal{M} \otimes S, (w, \sigma) \models K_j\theta$ iff $\mathcal{M}, w \models K_j\theta$. In particular, if $K_j\chi$ fails before the signal, $E_G\chi$ fails after it.

Proof. τ_j is Oblivious everywhere, so $R'_j((w, \sigma)) = \{(v, n) : wR_jv\}$, every (v, n) is in W' , and $V'(v, n) = V(v)$. $\square$

Proposition 3 is the reason the viewpoints are part of the floor and not of the robots' knowledge: what makes the announcement public is that each robot's view of the beacon is itself common knowledge. In the domain *at-view* is set by the public *go-view*, so once both robots have moved its value is common knowledge, and the hypothesis holds. Proposition 4 is checked by the fourth case of the validator: the beacon lit with north at no viewpoint leaves the depth at E^0 , with three worlds, and *lift* inapplicable.

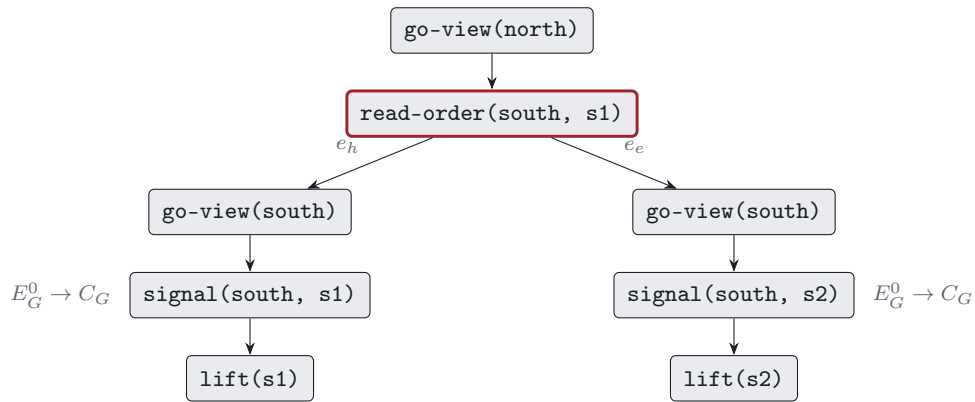


Figure 4: The policy on the beacon floor: depth five, two leaves. The radio is in the domain and is not used. Before the branch both leaves prescribe the same actions; after it the branch is announced on the beacon, so neither robot needs to know which branch it is in to execute its part.

6 Positions that are not announced

Proposition 3 needs the positions of the robots to be common knowledge, and the domain of Section 3 supplies that by making the move to a viewpoint public. The domain `coordinated-attack-positions` removes the assumption. A move is witnessed only by the robot that makes it, the signal can be made without its listener watching, and the floor offers one further action: from their viewpoints in t_2 the robots see each other. Two instances differ in one line, whether that sight line exists.

Definition 8 (Unwitnessed move). $U(i)$ has events do , with the move's precondition and postcondition, and $skip$, with $pre = \top$ and no effect; $E_d = \{do\}$; $Q(\tau_i)$ is the identity and $Q(\tau_j) = \{do, skip\}^2$ for the other agent j .

The other robot considers possible that the move did not happen. It does not believe that it did not, as an oblivious agent of a private ontic action would: no robot holds a false belief about the other's position, and the model stays S5.

Definition 9 (Unconfirmed announcement). $A(i, j, \chi)$ has events σ with $pre(\sigma) = at-view(i) \wedge at-view(j) \wedge \chi$, σ' with $pre(\sigma') = at-view(i) \wedge \neg at-view(j) \wedge \chi$, and n with $pre(n) = \top$; σ and σ' set $lit(s)$; $E_d = \{\sigma, \sigma'\}$; $Q(\tau_i) = \{\sigma, \sigma'\}^2 \cup \{(n, n)\}$ and $Q(\tau_j) = \{(\sigma, \sigma)\} \cup \{\sigma', n\}^2$.

It is the lossy message of Definition 6, with the listener's position deciding which of the first two events occurs. By Remark 1, a conditional audience would have made the listener's observation common knowledge regardless of where it stands; here it is not. The sighting is $S(\cdot, \top)$ with precondition $sightline \wedge at-view(i) \wedge at-view(j)$: an announcement that both robots stand at their viewpoints, to whoever stands at one.

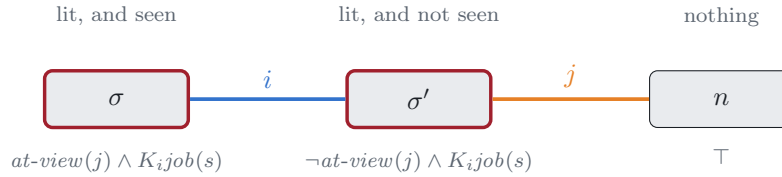


Figure 5: The unconfirmed announcement $A(i, j, K_i job(s))$. Both σ and σ' are designated, and the listener's position decides which can occur.

Proposition 5 (A signal to a robot that may not be watching). Let $\mathcal{M}$ be S5 with $\mathcal{M} \models \varphi$, let w be designated with $\mathcal{M}, w \models at-view(i) \wedge at-view(j) \wedge K_i \varphi$, and let $wR_i u$ with $\mathcal{M}, u \models \neg at-view(j)$, and $uR_j v$ with $\mathcal{M}, v \not\models \varphi$. Then $\text{dist}_{\mathcal{M} \otimes A(i, j, K_i \varphi)}(\varphi) \leq 2$: after the signal $E_G^2 \varphi$ fails.

Proof. $at-view(i)$ holds at u , since i witnesses its own moves and so knows its position, and $K_i \varphi$ holds at u by positive introspection; hence $(u, \sigma') \in W'$. Then $(w, \sigma) R'_i (u, \sigma')$ by $wR_i u$ and $(\sigma, \sigma') \in Q(\tau_i)$, and $(u, \sigma') R'_j (v, n)$ by $uR_j v$ and $(\sigma', n) \in Q(\tau_j)$, with $pre(n) = \top$; and (v, n) falsifies φ , as n changes nothing. $\square$

On the floor, after $go-view(north)$, $read-order(south, s_1)$ and $go-view(south)$, the hypotheses hold with u the copy in which north skipped its move: south cannot exclude that north never left its start, and in that world north does not know the order. The signal then gives $E_G^1 job(s_1)$, as one radio message does, and no more.

Proposition 6 (The sighting). If $at-view(i) \wedge at-view(j)$ holds at the designated worlds, the sighting yields $C_G(at-view(i) \wedge at-view(j))$. If it holds before $A(i, j, K_i \chi)$, the signal yields $C_G \chi$; and if the sighting follows the signal, it also yields $C_G \chi$.

Proof. Both robots are at their viewpoints at the designated worlds, so both types of the sighting are Fully, and it is a public announcement of $at-view(i) \wedge at-view(j)$: every world reachable

afterwards satisfies it, which is the first claim. If the positions are common knowledge before the signal, no world reachable from the designated (w, σ) satisfies $\neg at-view(j)$, so none of the form (u, σ') is reachable, the listener's relation keeps the σ -copies apart from the n -copies, and every reachable world is a σ -copy, at which $K_i\chi$, hence χ , held. If the sighting comes second, its precondition excludes every σ' -copy, in which $at-view(j)$ fails, and with them the n -copies, which were reachable only through a σ' -copy; what remains reachable is again σ -copies. $\square$

The last case says that a sighting makes an earlier signal common knowledge after the fact: in every world in which both robots stood at their viewpoints, the signal was seen. Without the sight line nothing in the domain announces a position, the copies in which the listener did not move stay reachable through every action, and by Propositions 5 and 2 the distance to a counterexample stays finite: $C_{Gjob}(s)$ never holds.

Table 3: The positions domain: 15 atoms and 30 ground actions on each floor, and the depth of $job(s_1)$ along three orders, from the trace.

instance or order	result
viewpoints see each other	policy, depth 6, 2 leaves, 298 expansions
viewpoints do not	no policy, space exhausted at depth 8
signal, no sighting	E^1 , <i>lift</i> not applicable
sighting, then signal	C
signal, then sighting	C

The policy is *go-view(north)*, *read-order(south, s₁)*, then on each outcome *go-view(south)*, the sighting, the signal for the stand, and the lift. The planner places the sighting before the signal; by Proposition 6 the reverse order would serve as well. Both instances were run in the simulator; Section 9 reports the runs.

7 Planning results

plank grounds each floor into 12 atoms and 28 ground actions, with the intermediate library and the lossy library together. Aletheia, through the ePlanSys plan solver, gives the results of Table 4.

Table 4: Search results. The unbounded variant is the radio floor of an earlier version of the domain without the send-once log, run with a 240s budget.

instance	result	measure
beacon floor	policy, depth 5, 2 leaves	115 expansions, 154 generated, 0.2 s
radio floor, one message per level	no policy	space exhausted at depth 5, 0.2 s
radio floor, messages repeatable	no answer	searched to depth 8, timed out at 9

Remark 2. The exhaustion on the bounded radio floor proves the bounded statement; the timeout on the unbounded one proves nothing; Corollary 1 proves the unbounded one. The three agree, and only the corollary is general.

8 Grounding on the floor

The floor is the pass-through floor: a racking block crosses a 30×50 m hall from wall to wall, and its three bays are the only gaps in it. The loads in t_1 and t_3 are the stands s_1 and s_2 ; the beacon stands in the open bay t_2 , and its two mouths are the viewpoints (Figure 6).

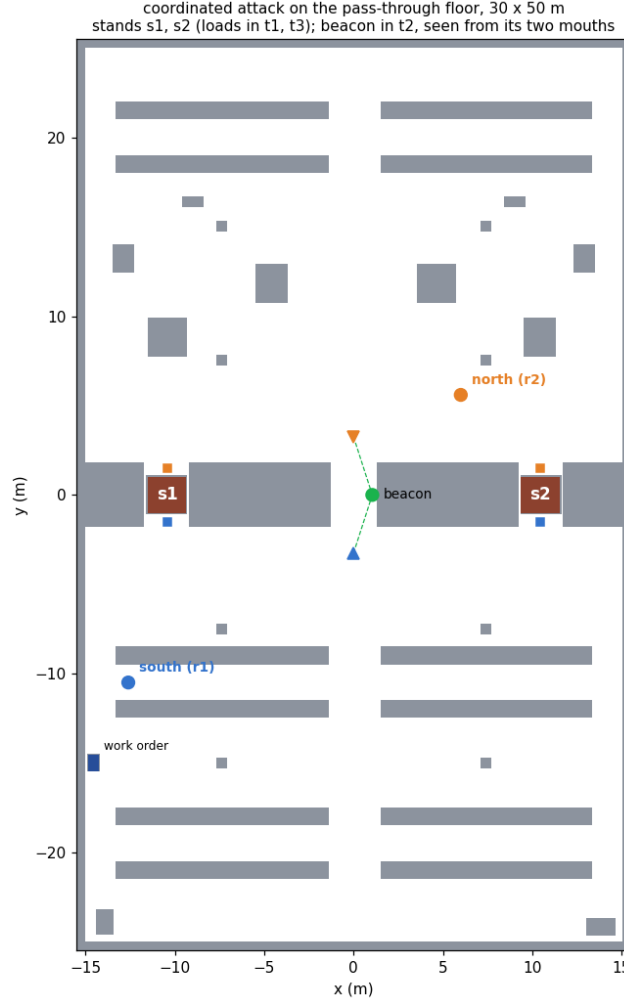


Figure 6: The floor plan both robots hold. The stands, the beacon and its two sight lines, the work-order terminal, and where each robot stands under each load.

The domain makes four claims about the geometry, and the floor generator checks each on the plan, by casting rays over the occupied cells, before it writes anything (Table 5). The third is the one the argument depends on: two robots under the two ends of one load are three metres apart and cannot see each other, so meeting at the stand gives no more common knowledge than the start did.

Table 5: The geometric claims, each checked on the floor plan.

claim	what depends on it
from where they start, neither robot sees the other or the beacon	the radio is the only channel on the radio floor
from its viewpoint, each robot sees the beacon	the observability condition of <i>signal</i>
under the two ends of one load, the robots do not see each other	meeting at the stand does not replace the beacon
every place a robot is sent is reachable over the inflated plan	a correct policy is drivable

Each action is performed by a ROS 2 node, and the model is advanced by the ePlanSys epistemic state after each one, with Aletheia’s own product update. The executor checks every action’s knowledge requirement against the model before it dispatches the action; for *lift(s₁)* that requirement is (C (south north) job_s1).

Table 6: How each action is performed.

action	performer
<i>read-order</i>	south drives to the terminal; the outcome is the work order, given to the node as the warehouse's record
messages	the message is put on the receiver's inbox and its arrival awaited, which checks that the designated event is the one that occurred; nothing reports it to the sender
<i>go-view</i>	the robot drives to its mouth of t_2 , and the beacon is checked to be in line of sight from where it stopped
<i>signal</i>	the beacon's tier for the stand is lit, after checking that every robot the model places at a viewpoint has the beacon in line of sight; the node refuses to signal if one does not
<i>lift</i>	both robots drive to their mouths of the stand, one start time is fixed for both, each drives in under its end, and the load is raised

9 Runs

The four floors were recorded with the order naming s_1 , on a 3840×1080 virtual display with RViz on the left and Gazebo on the right, and composed into two films, the radio and beacon floors in one and the two positions floors in the other, whose captions are the lines the runs wrote. The second branch of the beacon floor was run without recording. In the tables the model column gives worlds and designated worlds after the update, and the depth of mutual knowledge of $job(s_1)$.

9.1 The radio floor

Table 7: The radio floor, as the run reported it.

action	reported	model
plan	no policy for <i>lifted</i> after 0.2 s; space exhausted at depth 5	2 / 2
<i>read-order</i>	at the terminal after 12 s; the order names s_1 ; e_h	2 / 1, E^0
<i>tell</i>	" $K_{south}job(s_1)$ " delivered	4 / 1, E^1
<i>ack</i>	" $K_{north}K_{south}job(s_1)$ " delivered	6 / 1, E^2
<i>ack2</i>	delivered	8 / 1, E^3
<i>ack3</i>	delivered	10 / 1, E^4
<i>lift(s_1)</i>	knowledge requirement does not hold: (C (south north) job_s1)	refused

The chains computed from the live model after each message were those of Table 2, and the mission confirmed through the epistemic state that $K_{north}K_{south}K_{north}K_{south}job(s_1)$ held and $C_Gjob(s_1)$ did not. Neither robot moved toward a stand.

9.2 The beacon floor

Table 8: The beacon floor, as the run reported it.

action	reported	model
plan	8 policy items, 2 leaves, 0.2 s	2 / 2
<i>go-view(north)</i>	at its viewpoint after 17 s; beacon in line of sight, 3.5 m	2 / 2
<i>read-order</i>	at the terminal after 12 s; the order names s_1 ; e_h	2 / 1, E^0
<i>go-view(south)</i>	at its viewpoint after 47 s; beacon in line of sight, 3.5 m	2 / 1, E^0
<i>signal(south, s_1)</i>	in line of sight: south (3.5 m), north (3.5 m)	3 / 1, C
<i>lift(s_1)</i>	at the mouths after 22 and 23 s; joint start; under the load after 13.5 and 13.1 s: starts 0.00 s apart, arrivals 0.40 s apart; load raised 0.12 m	1 / 1, C

With the order naming s_2 the read gave e_e , the s_2 tier was lit, $C_G \text{job}(s_2)$ held, and the load on s_2 was raised with arrivals 0.60 s apart.

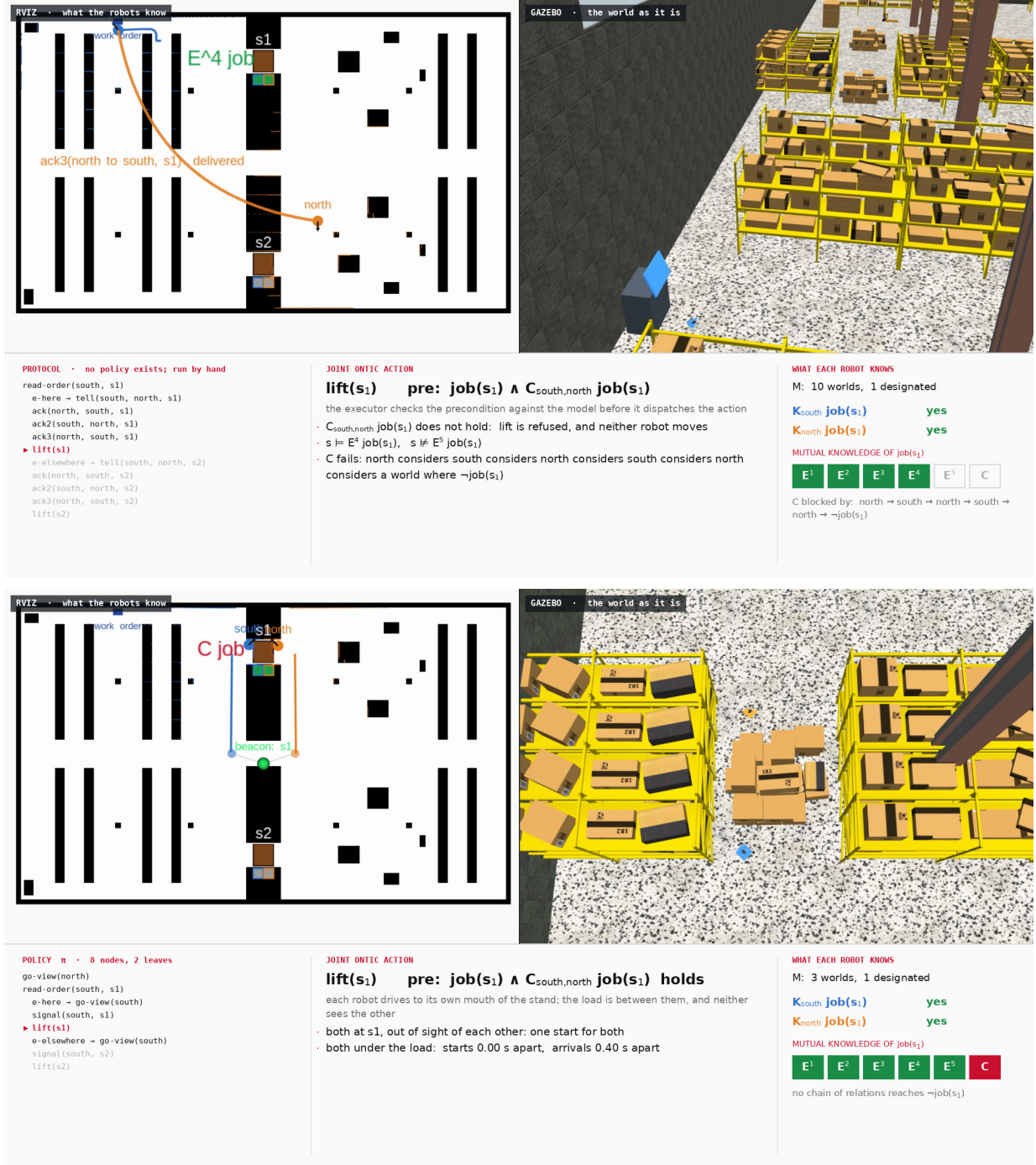


Figure 7: Two frames of the film. Above, the radio floor after the fourth message: E^4 , and the lift refused. Below, the beacon floor during the lift, seen from above the bay: C , both robots under the load.

9.3 The positions floors

Both instances of Section 6 were run: the sight instance on the beacon floor, and the blind instance on the same floor with a stack of crates on the axis of t_2 . The floor-plan check confirms that the crates leave the beacon in line of sight from each viewpoint and hide each viewpoint from the other. The sighting is performed by each robot's laser: along the bearing to the other robot, no beam may return short of it.

Table 9: The blind floor, as the run reported it.

action	reported	model
plan	no policy for <i>lifted</i> after 0.4s; space exhausted at depth 8	2 / 2
<i>go-view</i> (<i>north</i>)	at its viewpoint after 18s; beacon in line of sight, 3.5 m	4 / 2
<i>read-order</i>	at the terminal after 19s; the order names s_1 ; e_h	4 / 1, E^0
<i>go-view</i> (<i>south</i>)	at its viewpoint after 48s; beacon in line of sight, 3.5 m	8 / 1, E^0
<i>signal</i> (<i>south</i> , <i>north</i> , s_1)	in line of sight: south and north, 3.5 m; north sees it: e-signal-seen	10 / 1, E^1
<i>lift</i> (s_1)	knowledge requirement does not hold: (C (south north) job_s1)	refused

Table 10: The sight floor, as the run reported it.

action	reported	model
plan	10 policy items, 2 leaves, 0.2 s	2 / 2
<i>go-view</i> (<i>north</i>)	at its viewpoint after 17s; beacon in line of sight, 3.5 m	4 / 2
<i>read-order</i>	at the terminal after 13s; the order names s_1 ; e_h	4 / 1, E^0
<i>go-view</i> (<i>south</i>)	at its viewpoint after 50s; beacon in line of sight, 3.5 m	8 / 1, E^0
<i>sight</i> (<i>north</i> , <i>south</i>)	each robot 6.6 m from the other; each laser reads 6.7 m along the bearing: clear	10 / 1, E^0
<i>signal</i> (<i>south</i> , <i>north</i> , s_1)	north sees it: e-signal-seen	13 / 1, C
<i>lift</i> (s_1)	at the mouths after 23s each; joint start; under the load after 14.1 and 14.5 s: starts 0.00 s apart, arrivals 0.40 s apart; load raised 0.12 m	1 / 1, C

Each unwitnessed move doubled the model, and until the sighting neither robot knew where the other stood: $\neg K_{south} at-view(north)$ and $\neg K_{north} at-view(south)$ held on both floors. On the sight floor both became known with the sighting, and the signal then gave $C_G job(s_1)$. On the blind floor the signal gave north the stand, and also south's position, since the signal is made only from a viewpoint; south learned nothing about north. The chain that blocked C was $south \rightarrow north$: south considers a world in which north never left its start, and north there does not know the stand. The mission confirmed through the epistemic state that $K_{north} job(s_1)$ held and $K_{south} K_{north} job(s_1)$ did not, as Proposition 5 states.

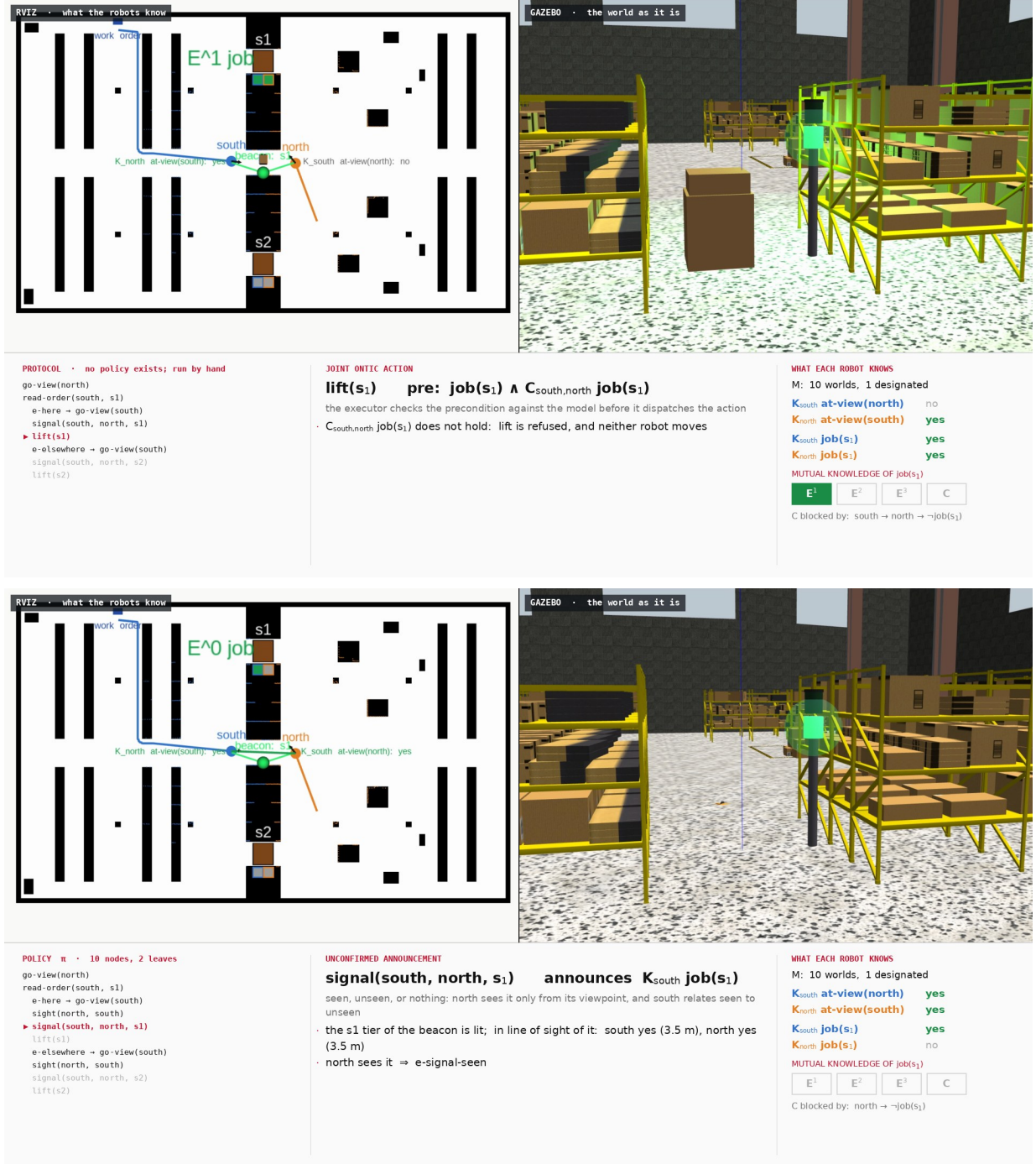


Figure 8: Two frames of the second film. Above, the blind floor at the refused lift: the crates in t_2 , E^1 , and south not knowing where north stands. Below, the sight floor at the signal: both positions known after the sighting.

10 Defects

One event named twice is one event. The lost event was first given by passing the delivered event a second time. plank keys an event model by event name, so the two became one, the loss disappeared from the model, and the radio floor was reported unsolvable at depth three for the wrong reason. Each message now has a lost event of its own.

A signal from the terminal. Before *read-order* required $\neg at-view(i)$, the policy sent south to its viewpoint and then to the terminal, and had it signal from there. The model placed south at

its viewpoint; the line-of-sight check read `south no`, `north yes`. The domain now orders the two, and the signal node refuses whenever the model and the floor disagree about who sees the beacon.

A static group modality. A common-knowledge modality over an atom no action changes requires `:static-common-knowledge`, and plank warns under `:common-knowledge`.

Teardown. In the runs of the radio and beacon floors the action nodes exited on signal 11 as the launch shut down, after the result was logged. The cause was a mismatch of two builds of `rcclcpp_cascade_lifecycle`: the executor's packages are compiled against the header ROS 2 Humble ships, and its workspace loads a newer library whose node class has two members more. The launch now preloads the library the headers describe, and in the runs of the positions floors every node exited cleanly.

11 What the runs establish

That common knowledge can be stated as the precondition of a joint action and enforced at execution: the executor evaluated it against the model the epistemic state maintained, refused the action on the floor where it could not hold, and dispatched it on the floor where it did.

That the lossy message, as an event model, adds at most one level of mutual knowledge per message whether or not it is delivered, so that on the radio floor no policy exists for any number of messages; and that the bound is attained on the run, four delivered messages yielding E^4 and no more.

And that a signal yields common knowledge exactly when each robot's observation of it is itself common knowledge, which the floor supplies through two viewpoints and the domain through a public move to them. When the move is not public, the runs show the same condition established on the floor: a sighting made with each robot's own laser preceded the signal on one floor and the lift was executed, and on the floor without a sight line the same signal gave E^1 and the lift was refused.

12 Directions for the next iteration

Positions without a sight line. Section 6 restores common knowledge of the robots' positions by a sighting, and on the floor without a sight line no policy exists. The radio cannot replace the sighting, by Corollary 1. A third observer that both robots are known to consult, such as a ceiling camera publishing positions to both, would be a public source of positions, and the natural next floor.

Probabilistic coordination. Common knowledge is unattainable over a lossy channel; coordination with a bounded probability of failure is not. A version in which each robot raises its end after a number of acknowledgements chosen against a loss rate would trade the precondition for a risk, and would need the belief layer to state it.

A channel that does fail. The link-outage apparatus gates the topics a message uses. Composing it with this domain would make delivery an observed outcome, with both d and l designated and the policy branching on what the receiver reports.

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Reproduction

```
ros2 launch coordinated_attack_demo coordinated_attack_launch.py floor:=radio
ros2 launch coordinated_attack_demo coordinated_attack_launch.py floor:=beacon
ros2 launch coordinated_attack_demo coordinated_attack_launch.py order:=s2
ros2 launch coordinated_attack_demo coordinated_attack_launch.py floor:=sight
ros2 launch coordinated_attack_demo coordinated_attack_launch.py floor:=blind
bash tools/validate.sh # eight checks, no simulator
bash tools/record_demo.sh radio s1 /tmp/raw_radio.mkv /tmp/run_radio.log
bash tools/record_demo.sh beacon s1 /tmp/raw_beacon.mkv /tmp/run_beacon.log
bash tools/record_demo.sh sight s1 /tmp/raw_sight.mkv /tmp/run_sight.log
bash tools/record_demo.sh blind s1 /tmp/raw_blind.mkv /tmp/run_blind.log
```